

Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

Win condition: Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

THE SCIENCE YOU ARE USING

Drag, lift, and hang time. A maple samara spins as it falls. The wing generates lift that slows the descent, so even a light breeze carries the seed away from the shade of its parent. More surface area for the same mass usually means a longer, slower fall.

One variable at a time. Wing length, wing angle, and added mass all change flight. Engineers learn what each does by changing only one and comparing. Copying a natural strategy on purpose is biomimicry.

HOW TO BUILD AND RUN IT

- 01 Study a real samara.** Drop a maple seed or a model. Watch how the wing makes it spin and slow down.
- 02 Build a baseline seed.** Make a simple winged seed with one paper clip for mass. Record a baseline drop time and distance.
- 03 Change one variable.** Adjust wing length OR mass OR fold angle, only one. Predict the effect before testing.
- 04 Test for hang time and distance.** Drop from the standard height or release in the fan lane. Time three trials and average.
- 05 Aim for a target.** In a final round, try to land in a marked zone. Control plus distance both score.

Safety: No throwing at people. Use marked launch lanes and low-power fan settings. Allergy: None.

MATERIALS

Paper and cardstock	shared
Paper clips	for mass
Masking tape	shared
String	shared
Fan or marked lane	1 test zone
Timers	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	100